P4.10 Start with $|1,2\rangle$ basis states (p = electric dipole moment, $\mathcal{E} = \mathcal{E}(t)$ el. field)

$$H_{ij} \to H = \begin{pmatrix} E_0 + p\mathcal{E} & -A \\ -A & E_0 - p\mathcal{E} \end{pmatrix}$$

$$|I\rangle = \frac{1}{\sqrt{2}}|1\rangle + \frac{1}{\sqrt{2}}|2\rangle \; ; \; |II\rangle = \frac{1}{\sqrt{2}}|1\rangle - \frac{1}{\sqrt{2}}|2\rangle$$

$$H_{KM} = \langle K|\hat{H}|M\rangle = \sum_{ij} \langle K|i\rangle\langle i|\hat{H}|j\rangle\langle j|M\rangle$$

$$= S^+_{Ki} H_{ij} S_{jM}$$

$$S = \begin{pmatrix} \langle 1|I\rangle & \langle 1|II\rangle \\ \langle 2|I\rangle & \langle 2|II\rangle \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = S^+$$

$\Rightarrow$ in $|I\rangle, |II\rangle$ basis

$$H \to \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} E_0 + p\mathcal{E} & -A \\ -A & E_0 - p\mathcal{E} \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} =$$

$$= \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} E_0 + p\mathcal{E} - A & E_0 + p\mathcal{E} + A \\ -A + E_0 - p\mathcal{E} & -A - E_0 + p\mathcal{E} \end{pmatrix} =$$

$$= \frac{1}{2}\begin{pmatrix} 2(E_0 - A) & 2p\mathcal{E} \\ 2p\mathcal{E} & 2(E_0 + A) \end{pmatrix} = \begin{pmatrix} E_0 - A & p\mathcal{E} \\ p\mathcal{E} & E_0 + A \end{pmatrix}$$

Take $\mathcal{E} = \mathcal{E}_0 \cos\omega t$ and $|\psi\rangle = a|I\rangle + b|II\rangle$

Then S.E. $\quad i\hbar \frac{d}{dt}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} E_0 - A & p\mathcal{E}_0 \cos\omega t \\ p\mathcal{E}_0 \cos\omega t & E_0 + A \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix}$

To bring it to the form (4.38) we need to get rid of E_0. This is done by substitution

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} e^{-iE_0 t/\hbar}$$

(this is just a shift of origin in E-space)

Indeed $i\hbar \frac{d}{dt}\begin{pmatrix} a \\ b \end{pmatrix} = i\hbar \frac{d}{dt}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} e^{-iE_0 t/\hbar} =$

$$= i\hbar \left\{ e^{-iE_0 t/\hbar} \frac{d}{dt}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \left(-\frac{iE_0}{\hbar}\right) e^{-iE_0 t/\hbar} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \right\}$$

Then S.E. becomes

$$i\hbar\, e^{-iE_0 t/\hbar} \frac{d}{dt}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \cancel{E_0 e^{-iE_0 t/\hbar} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}} =$$

$$= \cancel{E_0 \begin{pmatrix} \alpha \\ \beta \end{pmatrix} e^{-iE_0 t/\hbar}} + \begin{pmatrix} -A & p\mathcal{E}_0 \cos\omega t \\ p\mathcal{E}_0 \cos\omega t & A \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} e^{-iE_0 t/\hbar}$$

Divide by $e^{-iE_0 t/\hbar}$ now:

$$i\hbar \frac{d}{dt}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} -A & p\mathcal{E}_0 \cos\omega t \\ p\mathcal{E}_0 \cos\omega t & A \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

Can obtain form (4.38) by transcription:

$$-A \to \frac{\hbar\omega_0}{2}$$

$$p\mathcal{E}_0 \to \frac{\hbar\omega_1}{2}$$

Then $|\langle \mathrm{II} | \psi(t) \rangle|^2 = |b(t)|^2 = |\beta e^{-iE_0 t/\hbar}|^2 = |\beta|^2$

Rabi's f-la $|\langle -z | \psi(t) \rangle|^2 = \dfrac{\omega_1^2/4}{(\omega_0-\omega)^2 + \frac{\omega_1^2}{4}} \sin^2 \frac{1}{2}\sqrt{(\omega_0-\omega)^2 + \frac{\omega_1^2}{4}}\, t$

$$= \frac{\left(\frac{\hbar\omega_1}{2}\right)^2}{(\hbar\omega_0 - \hbar\omega)^2 + \left(\frac{\hbar\omega_1}{2}\right)^2} \sin^2 \frac{1}{2\hbar}\sqrt{(\hbar\omega_0 - \hbar\omega)^2 + \left(\frac{\hbar\omega_1}{2}\right)^2}\, t \longrightarrow$$

$$\longrightarrow \frac{(p\mathcal{E}_0)^2}{(2A - \hbar\omega)^2 + (p\mathcal{E}_0)^2} \sin^2\left[\frac{1}{2\hbar}\sqrt{(2A-\hbar\omega)^2 + (p\mathcal{E}_0)^2}\, t\right]$$

At the resonance $\hbar\omega = 2A$ = energy difference $(E_{II} - E_{I})$

$$|\langle II|\psi(t)\rangle|^2 = |\beta|^2 = \sin^2\left(\frac{p\mathcal{E}_0 t}{2\hbar}\right).$$ This is (4.62).

<u>Note</u>: it appears that since $\hbar\omega_0 \to -2A$, then $(\omega_0-\omega)^2$ in Rabi's f-la becomes

$$(-2A-\hbar\omega)^2 = (2A+\hbar\omega)^2.$$

But this is wrong for 2 reasons:

① Physically we know that the resonance is possible for <u>positive</u> frequency $\omega = \frac{2A}{\hbar}$

This requires $(2A-\hbar\omega)^2$ in equation for $|\beta|^2$.

② When deriving Rabi's f-la, we kept "slow" oscillations $e^{i(\omega_0-\omega)t}$ and neglected "fast" ones $e^{i(\omega_0+\omega)t}$.

With $\omega_0 \to -\frac{2A}{\hbar}$ replacement "fast" and "slow" interchange: <u>slow</u> $e^{i(\omega_0-\omega)t} \to e^{-i(2A+\hbar\omega)t/\hbar}$ <u>fast</u>

<u>fast</u> $e^{i(\omega_0+\omega)t} \to e^{-i(2A-\hbar\omega)t/\hbar}$ <u>slow</u>

$\Rightarrow$ we do need to keep only $e^{\pm i(2A-\hbar\omega)t/\hbar}$ oscillations and this leads to

$$|\beta|^2 = \frac{(p\mathcal{E}_0)^2}{(p\mathcal{E}_0)^2+(2A-\hbar\omega)^2}\sin^2\left[\frac{1}{2\hbar}\sqrt{(2A-\hbar\omega)^2+(p\mathcal{E}_0)^2}\,t\right]$$

$\boxed{P4.13}$

$$H = \begin{pmatrix} E_0 & 0 & A \\ 0 & E_1 & 0 \\ A & 0 & E_0 \end{pmatrix}$$ for a 3-state system in the basis $|1\rangle, |2\rangle, |3\rangle$.

Need to find eigenvalues and eigenstates:

$$H \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} E_0 & 0 & A \\ 0 & E_1 & 0 \\ A & 0 & E_0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \lambda \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

Eigenvalues follow from

$$\det \begin{pmatrix} E_0 - \lambda & 0 & A \\ 0 & E_1 - \lambda & 0 \\ A & 0 & E_0 - \lambda \end{pmatrix} = (E_1 - \lambda)\left((E_0 - \lambda)^2 - A^2\right) = 0$$

$$\Rightarrow \lambda_1 = E_1 \quad , \quad \lambda_{2,3} = E_0 \pm A$$

Eigen[redacted] vectors are:

$$\boxed{\lambda_1 = E_1} \Rightarrow \begin{cases} (E_0 - E_1)\, a + A c = 0 \\ A a + (E_0 - E_1)\, c = 0 \end{cases}$$

$$\Rightarrow a = -\frac{E_0 - E_1}{A} c \Rightarrow -\frac{(E_0 - E_1)^2}{A} c + A c = 0$$

$$\Rightarrow c \left[A - \frac{(E_0 - E_1)^2}{A} \right] = 0 \Rightarrow \begin{cases} c = 0 \\ a = 0 \end{cases}$$

$$\Rightarrow |\psi_1\rangle = \begin{pmatrix} 0 \\ b \\ 0 \end{pmatrix} \Rightarrow \text{normalization requires } b = 1$$

$$\Rightarrow |\psi_1\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = |2\rangle .$$

(a) $$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |2\rangle = e^{-iE_1 t/\hbar} |\psi_1\rangle =$$

$$= e^{-iE_1 t/\hbar} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} .$$

(b) For $\lambda_2 = E_0 + A$ eigenvector is

$$\begin{pmatrix} -A & 0 & A \\ 0 & E_1 - E_0 - A & 0 \\ A & 0 & -A \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0$$

$$\Rightarrow \begin{cases} -A(a-c) = 0 \\ (E_1 - E_0 - A) b = 0 \\ A(a-c) = 0 \end{cases} \Rightarrow \begin{cases} a = c \\ b = 0 \\ a = c \end{cases}$$

$$\Rightarrow |\psi_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

for $\lambda_3 = E_0 - A$ $\quad |\psi_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ (check it!)

Thus $|\psi_2\rangle = \frac{1}{\sqrt{2}} |1\rangle + \frac{1}{\sqrt{2}} |3\rangle$

$$|\psi_3\rangle = \frac{1}{\sqrt{2}} |1\rangle - \frac{1}{\sqrt{2}} |3\rangle$$

$$\Rightarrow |3\rangle = \frac{1}{\sqrt{2}} |\psi_2\rangle - \frac{1}{\sqrt{2}} |\psi_3\rangle$$

$$\Rightarrow |\psi(t)\rangle = e^{-i\hat{H}t/\hbar} \frac{1}{\sqrt{2}} \left(|\psi_2\rangle - |\psi_3\rangle \right) =$$

$$= \frac{e^{-i\lambda_2 t/\hbar}}{\sqrt{2}} |\psi_2\rangle - \frac{e^{-i\lambda_3 t/\hbar}}{\sqrt{2}} |\psi_3\rangle =$$

$$= \frac{1}{\sqrt{2}} e^{-i(E_0 + A)t/\hbar} |\psi_2\rangle - \frac{1}{\sqrt{2}} e^{-i(E_0 - A)t/\hbar} |\psi_3\rangle$$

$$= \frac{1}{\sqrt{2}} e^{-iE_0 t/\hbar} \left(e^{-iAt/\hbar} |\psi_2\rangle - e^{iAt/\hbar} |\psi_3\rangle \right)$$

$\Rightarrow$ in the $|1,2,3\rangle$ basis

$$|\psi(t)\rangle = \frac{e^{-iE_0t/\hbar}}{\sqrt{2}}\left\{\frac{e^{-iAt/\hbar}}{\sqrt{2}}\begin{pmatrix}1\\0\\1\end{pmatrix} - \frac{e^{iAt/\hbar}}{\sqrt{2}}\begin{pmatrix}1\\0\\-1\end{pmatrix}\right\}$$

$$= \frac{e^{-iE_0t/\hbar}}{2}\begin{pmatrix}e^{-iAt/\hbar} - e^{iAt/\hbar}\\ 0\\ e^{-iAt/\hbar} + e^{iAt/\hbar}\end{pmatrix} =$$

$$= e^{-iE_0t/\hbar}\begin{pmatrix}-i\sin\left(\frac{At}{\hbar}\right)\\ 0\\ \cos\left(\frac{At}{\hbar}\right)\end{pmatrix}$$

while initial state in this basis was $|3\rangle = \begin{pmatrix}0\\0\\1\end{pmatrix}$.

$\boxed{P4.14}$ $\hat{H} \underset{\substack{|x\rangle|y\rangle \\ \text{basis}}}{\rightarrow} \begin{pmatrix}0 & -iE_0\\ iE_0 & 0\end{pmatrix}$

(a) $\begin{pmatrix}0 & -iE_0\\ iE_0 & 0\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix} = E\begin{pmatrix}a\\b\end{pmatrix}$

$$\det\begin{pmatrix}-E & -iE_0\\ iE_0 & -E\end{pmatrix} = E^2 - (iE_0)(-iE_0) = E^2 - E_0^2 = 0$$

$E_{1,2} = \pm E_0$

E_1 $\begin{pmatrix}0 & -iE_0\\ iE_0 & 0\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix} = E_0\begin{pmatrix}a\\b\end{pmatrix} \Rightarrow \begin{cases}-iE_0 b = E_0 a\\ iE_0 a = E_0 b\end{cases}$

$\Rightarrow a = -ib \Rightarrow |\psi_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}-i\\1\end{pmatrix} = \frac{-i}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix}$

(E₂) $$\begin{pmatrix} 0 & -iE_0 \\ iE_0 & 0 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = -E_0 \begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow \begin{cases} -iE_0 b = -E_0 a \\ iE_0 a = -E_0 b \end{cases}$$

$$\Rightarrow \underline{a = ib} \quad \Rightarrow |\psi_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} i \\ 1 \end{pmatrix} = \frac{i}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Comparing with (2.111) we see that (upto a phase)

$$\left.\begin{aligned} |\psi_1\rangle &= |R\rangle = \frac{1}{\sqrt{2}}\left(|X\rangle + i|Y\rangle\right) \\ |\psi_2\rangle &= |L\rangle = \frac{1}{\sqrt{2}}\left(|X\rangle - i|Y\rangle\right) \end{aligned}\right\} \begin{aligned} &E_1 = E_R = E_0 \\ &\text{circularly polarized} \\ &\text{photon states} \\ &E_2 = E_L = -E_0 \end{aligned}$$

(b) $$|\psi(0)\rangle = |X\rangle = \frac{1}{\sqrt{2}}\left(|R\rangle + |L\rangle\right)$$

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|X\rangle = \frac{e^{-iE_0 t/\hbar}}{\sqrt{2}}|R\rangle + \frac{e^{iE_0 t/\hbar}}{\sqrt{2}}|L\rangle$$

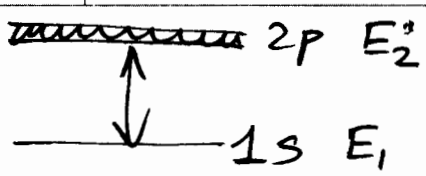
$$= \frac{1}{\sqrt{2}}\left\{e^{-iE_0 t/\hbar}\frac{1}{\sqrt{2}}\left(|X\rangle + i|Y\rangle\right) + e^{iE_0 t/\hbar}\frac{1}{\sqrt{2}}\left(|X\rangle - i|Y\rangle\right)\right\}$$

$$= \frac{1}{2}\left\{\left(e^{-iE_0 t/\hbar} + e^{iE_0 t/\hbar}\right)|X\rangle + \left(i\,e^{-iE_0 t/\hbar} - i e^{iE_0 t/\hbar}\right)\cdot|Y\rangle\right\}$$

$$= \cos\left(\frac{E_0 t}{\hbar}\right)|X\rangle + \sin\left(\frac{E_0 t}{\hbar}\right)|Y\rangle$$

This is linearly polarized photon.
Its direction of polarization $\phi = \frac{E_0 t}{\hbar}$ rotates with time.

$\boxed{P4.16}$ $\tau = 1.6 \cdot 10^{-9} s$



Using p.463

$$\Delta E = \frac{\hbar}{\tau} = \frac{1.05 \cdot 10^{-34} J.s}{1.6 \cdot 10^{-9} s} = 0.6 \cdot 10^{-25} J = \frac{0.6 \cdot 10^{-25}}{1.6 \cdot 10^{-19}} eV$$

$$1 J = \frac{1 eV}{1.6 \cdot 10^{-19}}$$

$$= 0.4 \cdot 10^{-6} eV$$

$$= 4 \cdot 10^{-7} eV$$

For infinitely sharp 2p level (so that $\Delta E = 0$) the wavelength of $1s \rightarrow 2p$ transition is found from $E_2 - E_1 = \frac{hc}{\lambda}$

$$\lambda = \frac{hc}{E_2 - E_1}$$

When $\Delta E \neq 0$, then $\lambda \pm \Delta\lambda = \frac{hc}{E_2 \pm \Delta E - E_1} =$

$$= \frac{hc}{(E_2 - E_1)\left(1 \pm \frac{\Delta E}{E_2 - E_1}\right)} = \frac{hc}{E_2 - E_1}\left(1 \mp \frac{\Delta E}{E_2 - E_1}\right) =$$

$$= \lambda \left(1 \mp \frac{\Delta E}{E_2 - E_1}\right)$$

$$\Rightarrow \pm \Delta\lambda = \mp \lambda \frac{\Delta E}{E_2 - E_1}$$

$$\left|\frac{\Delta\lambda}{\lambda}\right| = \frac{\Delta E}{E_2 - E_1}$$

$\Delta\lambda = 5 \cdot 10^{-5} Å$ is very small. Remember that "size of hydrogen atom" $\sim 1 Å$

Now, for Hydrogen $E_n = -\frac{13.6 eV}{n^2} \Rightarrow E_2 - E_1 = (-13.6 eV)\left(\frac{1}{2^2} - 1\right)$
$= 10.2 eV$

$$\Rightarrow \frac{\Delta\lambda}{\lambda} = \frac{4 \cdot 10^{-7} eV}{10 eV} = 4 \cdot 10^{-8} \Rightarrow \Delta\lambda = (1200 Å)(4 \cdot 10^{-8}) = 5 \cdot 10^{-5} Å$$